

1 Should site be a covariate in the analysis model?

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34 1 Abstract

35 **Background.** Multicenter randomised clinical trials recruit patients across multiple sites,
36 introducing site as a potential source of outcome variability. Whether and how site should
37 enter the analysis model — as a fixed effect, a random effect, or as the basis for treatment-
38 by-site interaction testing — is a decision with consequences for power, type I error, and
39 the inferential target of the trial. Regulatory guidance from ICH E9, the FDA, and the
40 EMA recommend that stratification variables be included in the analysis model, but the
41 choice between fixed and random effects, and the inclusion or exclusion of treatment-by-site
42 interaction terms, has remained pragmatic.

43 **Methods.** We conduct a Monte Carlo simulation comparing four linear-mixed analysis mod-
44 els (no site, site as fixed effect, site as random effect, site-by-treatment interaction) across
45 four scenarios crossing equal-versus-unequal site sizes with the presence-versus-absence of true
46 treatment-by-site interaction. Each scenario is evaluated at $n_{\text{sim}} = 1,000$ replicates with bias,
47 empirical SE, model-based SE, power, coverage, and convergence rate reported.

48 **Results.** Including site as a random effect generally performs as well as or better than
49 the fixed-effects approach across all scenarios examined, consistent with¹ and². The random-
50 effects model avoids the loss of degrees of freedom inherent in the fixed-effects approach, which
51 becomes particularly problematic when the number of sites is large relative to the total sample
52 size, and is amplified when site sizes are unequal. When no true treatment-by-site interaction
53 exists, including the interaction term expends degrees of freedom without benefit and reduces
54 power. Ignoring site when it is a true source of outcome variability inflates residual variance
55 and reduces power.

56 **Conclusions.** Site should be included in the analysis model as a random effect when site is a
57 stratification variable or contributes to outcome variability. The treatment-by-site interaction
58 term should be included only when there is a strong a priori reason to expect treatment effect
59 heterogeneity across sites. The simulation evidence supports the regulatory recommendation
60 that stratification variables enter the analysis as covariates.

61 2 Introduction

62 We address a key question in the analysis of data from multi-site randomized clinical trials:
63 whether and how to include site (or centre) as a covariate in the analysis model. The question
64 is not merely academic. The choice of whether to adjust for site, and whether to treat it as a
65 fixed or random effect, can influence estimates of treatment effect, standard errors, confidence
66 interval width, and statistical power^{1,3}.

67 Multicenter trials are conducted to increase enrollment rates and improve the generalizability

68 of findings across clinical settings. However, the inclusion of multiple sites introduces a source
69 of variability that complicates the analysis. Patients within a site tend to be more similar
70 to one another than to patients at other sites, owing to shared clinical practices, patient
71 populations, and local standards of care. This within-site correlation, or clustering, violates
72 the independence assumption underlying many standard statistical methods³.

73 2.1 Regulatory context

74 Both international and national regulatory bodies have addressed the role of covariates, includ-
75 ing site, in the analysis of clinical trials. The ICH E9 guideline states that ‘factors on which
76 randomisation has been stratified should be accounted for later in the analysis’⁴. When ran-
77 domization is stratified by site, this implies that site should appear in the analysis model. The
78 United States Food and Drug Administration (FDA) 2023 guidance on covariate adjustment
79 reinforces this position, recommending that stratification variables be included in the primary
80 analysis model⁵. The European Medicines Agency (EMA) similarly advises that stratification
81 factors be incorporated unless stratification was performed for purely administrative reasons⁶.

82 2.2 The case for adjusting for site

83 The primary statistical argument for including site in the analysis model is that it can reduce
84 residual variance, thereby increasing the precision of treatment effect estimates and improv-
85 ing power⁷. When outcome rates or levels differ substantially across sites, whether because
86 of differences in patient populations, diagnostic practices, or supportive care, failing to ac-
87 count for this heterogeneity leaves variance unexplained in the error term. Adjustment for
88 site absorbs this between-site variability and can yield narrower confidence intervals for the
89 treatment effect^{1,2}.

90 A further argument concerns the validity of inference when randomization has been stratified
91 by site.⁸ demonstrated that stratified randomization induces correlation between treatment
92 groups within strata. If the analysis ignores this structure, standard errors are biased upward,
93 confidence intervals are too wide, and power is reduced.⁹ provided a clinical example in which
94 failure to adjust for stratification factors led to a false negative conclusion.

95 2.3 Fixed versus random effects

96 A central debate concerns whether site should enter the model as a fixed or random effect.
97 Fixed effects treat each site as a separate parameter to be estimated, making no assumptions
98 about the distribution of site effects but consuming degrees of freedom proportional to the
99 number of sites. Random effects assume site effects are drawn from a normal distribution,
100 estimating only the variance of that distribution and thereby requiring fewer parameters¹⁰.

101 ¹ conducted a simulation study comparing fixed and random centre effects for continuous
102 outcomes across 378 scenarios. Random effects performed as well as or better than fixed effects
103 in all scenarios, with notable advantages when the number of patients per site was small (10
104 or fewer) or when there was imbalance between treatment arms within sites.¹¹ extended these

findings to binary outcomes, comparing fixed effects, random effects, generalized estimating equations (GEEs), and Mantel-Haenszel methods.

The practical relevance of this comparison is underscored by 1, who reviewed 206 multicenter trials in four major medical journals and found that the median number of events per centre-treatment arm combination was just 3, with 50% of trials having fewer than 3 events per centre-treatment combination. Fixed-effects models require substantial data per site to perform well; in the majority of real trials, this condition is not met¹².

2.4 Treatment-by-site interaction

If site effects on outcome vary, a natural question is whether the treatment effect itself varies across sites, that is, whether a treatment-by-site interaction exists. A qualitative interaction, where the direction of the treatment effect reverses at some sites, has major clinical implications: it suggests that the treatment may be harmful for some patient populations¹³. However, tests for treatment-by-site interaction are typically underpowered because individual sites rarely enroll enough patients to estimate site-specific treatment effects precisely².

¹⁴ discussed strategies for handling treatment-by-centre interaction with binary outcomes, noting that random effects models smooth extreme site-specific estimates and provide more stable inference than fixed effects when some sites contribute sparse data.¹⁵ reviewed recent practice and confirmed that the choice of method for handling centre effects can materially affect trial conclusions.

2.5 Heterogeneous site sizes

Real multicenter trials rarely have equal enrollment across sites. Some sites may contribute hundreds of patients while others contribute fewer than ten. This heterogeneity in site size interacts with the choice of analytic method. Fixed-effects models can lead to patient exclusion when sites have observations in only one treatment arm (a common occurrence with small sites), whereas random-effects models borrow strength across sites and avoid this problem^{12,16}.³ provided an overview of the analytic challenges introduced by clustering, confounding, and effect modification in multicenter studies, emphasizing that the correlation of outcomes within sites must be addressed to obtain valid inference.

2.6 Present study

Despite the extensive methodological literature, uncertainty persists about when and how to adjust for site in practice. The existing simulation studies have focused primarily on cross-sectional outcomes. In many clinical trials, however, outcomes are measured longitudinally over multiple visits, introducing an additional layer of within-subject correlation that interacts with the between-site structure. We use Monte Carlo simulation to compare four analysis strategies for a longitudinal outcome in a multicenter trial: (1) ignoring site entirely, (2) including site as a fixed effect, (3) including site as a random effect, and (4) including a site-by-treatment interaction term. We evaluate these strategies under equal and unequal site

142 sizes, with and without true treatment-by-site interaction, assessing bias, power, coverage,
143 and convergence.

144 3 Methods

145 3.1 ADEMP structure

146 Following Morris, White, and Crowther¹⁷, the simulation is reported under the ADEMP
147 framework.

- 148 • **Aims.** Compare four strategies for incorporating site information in multicenter trial
149 analyses (no site adjustment, fixed site effects, random site effects, treatment-by-site
150 interaction) across scenarios crossing balanced vs imbalanced site sizes with and without
151 true treatment-by-site heterogeneity.
- 152 • **Data-generating mechanism.** Detailed in the next subsection.
- 153 • **Estimand.** The overall treatment effect δ ; when treatment-by-site variance is non-zero,
154 the site-size-weighted average of site-specific effects.
- 155 • **Methods.** Four linear mixed models implemented via `lme4::lmer` differing only in
156 how site is entered.
- 157 • **Performance measures.** Convergence rate, bias, empirical SE, MSE, mean model-
158 based SE, power at $\alpha = 0.05$, and coverage of 95% CIs, each reported with its Monte
159 Carlo SE per Morris Table 6.

160 3.2 Data generating model

161 We simulate data from a two-arm parallel-group multicenter trial with a continuous longitu-
162 dinal outcome measured at $T = 4$ equally spaced time points. The model for the outcome
163 Y_{ijk} of subject j at site i and time k is:

$$164 \quad Y_{ijk} = \alpha_i + (\beta + \gamma_i) \cdot \text{Trt}_{ij} + \delta \cdot t_k + u_j + \epsilon_{ijk}$$

165 where $\alpha_i \sim N(0, \sigma_s^2)$ is the random site intercept, β is the true treatment effect, $\gamma_i \sim N(0, \sigma_{s \times t}^2)$
166 is the random treatment-by-site interaction, $\delta = 0.05$ is a linear time slope, $u_j \sim N(0, 0.25)$
167 is a subject-level random intercept, and $\epsilon_{ijk} \sim N(0, \sigma_e^2)$ is residual error.

168 3.3 Simulation scenarios

169 We examine four scenarios defined by crossing two factors:

- 170 • **Site size:** equal (20 patients per site) versus unequal (exponentially distributed with
171 mean 20, minimum 4)
- 172 • **Treatment-by-site interaction:** absent ($\sigma_{s \times t}^2 = 0$) versus present ($\sigma_{s \times t}^2 = 0.15$)

173 In all scenarios, we set $K = 20$ sites, $\beta = 0.30$, $\sigma_s^2 = 0.25$, and $\sigma_e^2 = 1.0$. Within each site,
174 subjects are randomized 1:1 to treatment and control.

175 3.4 Analysis models

176 For each simulated dataset, we fit four linear mixed models using the `lme4` package¹⁸:

- 177 1. **No site:** $y \sim \text{trt} + \text{time} + (1|\text{subj})$
- 178 2. **Site fixed:** $y \sim \text{trt} + \text{time} + \text{site} + (1|\text{subj})$
- 179 3. **Site random:** $y \sim \text{trt} + \text{time} + (1|\text{site}) + (1|\text{subj})$
- 180 4. **Site $\times$ treatment:** $y \sim \text{trt}*\text{site} + \text{time} + (1|\text{subj})$

181 3.5 Performance metrics

182 For each model across 1,000 replications, we compute:

- 183 • **Bias:** mean estimated treatment effect minus true value ($\beta = 0.30$)
- 184 • **Empirical SE:** standard deviation of estimates across replications
- 185 • **Mean model SE:** average of model-based standard errors
- 186 • **Power:** proportion of replications with $p < 0.05$
- 187 • **Coverage:** proportion of nominal 95% confidence intervals containing the true value
- 188 • **Convergence rate:** proportion of replications in which the model converged

189 For coverage MCSE ≤ 0.7 pp at 0.95, $n_{\text{sim}} \geq 970$; $n_{\text{sim}} = 1000$ gives coverage MCSE ≈ 0.69
190 pp.

191 4 Results

192 4.1 Headline findings

193 We find that across four scenarios crossing balanced vs imbalanced site sizes with and without
194 true treatment-by-site heterogeneity, all four analysis models produced absolute bias below
195 0.002 (worst case: `no_site` under “Equal, with interaction”, MCSE 0.002). Coverage of 95%
196 confidence intervals across all combinations fell in [0.914, 0.983]. Mean power by model
197 (averaged across scenarios) was: `no_site` = 0.94, `site_fixed` = 0.97, `site_random` = 0.97,
198 `site_x_trt` = 0.97. All performance measures are reported with their Morris Table 6 Monte
199 Carlo SEs in the per-scenario tables below.

200 4.2 Scenario 1: Equal site sizes, no interaction

Table 1: Performance of four analysis models: equal site sizes, no treatment-by-site interaction.

Model	Bias	Emp. SE	Model SE	Power	Coverage	Conv.	N conv.
No site	0.001	0.072	0.086	0.973	0.982	1	1000
Site (fixed)	0.001	0.072	0.071	0.991	0.953	1	1000
Site (random)	0.001	0.072	0.071	0.991	0.953	1	1000
Site x Trt	0.001	0.072	0.071	0.991	0.953	1	1000

4.3 Scenario 2: Unequal site sizes, no interaction

Table 2: Performance of four analysis models: unequal site sizes, no treatment-by-site interaction.

Model	Bias	Emp. SE	Model SE	Power	Coverage	Conv.	N conv.
No site	0.001	0.069	0.084	0.967	0.983	1	1000
Site (fixed)	0.001	0.068	0.069	0.989	0.943	1	1000
Site (random)	0.001	0.068	0.069	0.989	0.943	1	1000
Site x Trt	0.001	0.068	0.069	0.988	0.942	1	1000

4.4 Scenario 3: Equal site sizes, with interaction

Table 3: Performance of four analysis models: equal site sizes, with treatment-by-site interaction.

Model	Bias	Emp. SE	Model SE	Power	Coverage	Conv.	N conv.
No site	0.002	0.079	0.087	0.954	0.966	1	1000
Site (fixed)	0.002	0.079	0.071	0.982	0.929	1	1000
Site (random)	0.002	0.079	0.071	0.982	0.929	1	1000
Site x Trt	0.002	0.079	0.071	0.983	0.928	1	1000

4.5 Scenario 4: Unequal site sizes, with interaction

Table 4: Performance of four analysis models: unequal site sizes, with treatment-by-site interaction.

Model	Bias	Emp. SE	Model SE	Power	Coverage	Conv.	N conv.
No site	0.001	0.098	0.102	0.858	0.952	1	1000
Site (fixed)	0.001	0.097	0.084	0.919	0.916	1	1000
Site (random)	0.001	0.097	0.084	0.920	0.916	1	1000
Site x Trt	0.001	0.097	0.084	0.918	0.914	1	1000

4.6 Comparison of bias and power across scenarios

5 Discussion

The simulation results illuminate several practical considerations for the inclusion of site in the analysis of multicenter trials with longitudinal outcomes, which we shall now take up in turn.

First, consistent with the findings of¹ and², including site as a random effect generally performs as well as or better than fixed effects across all scenarios examined. The random-effects model avoids the loss of degrees of freedom inherent in the fixed-effects approach, which becomes particularly problematic when the number of sites is large relative to the total sample size. This advantage is amplified when site sizes are unequal, as fixed-effects models can produce unstable estimates when some sites contribute very few subjects per treatment arm^{12,16}.

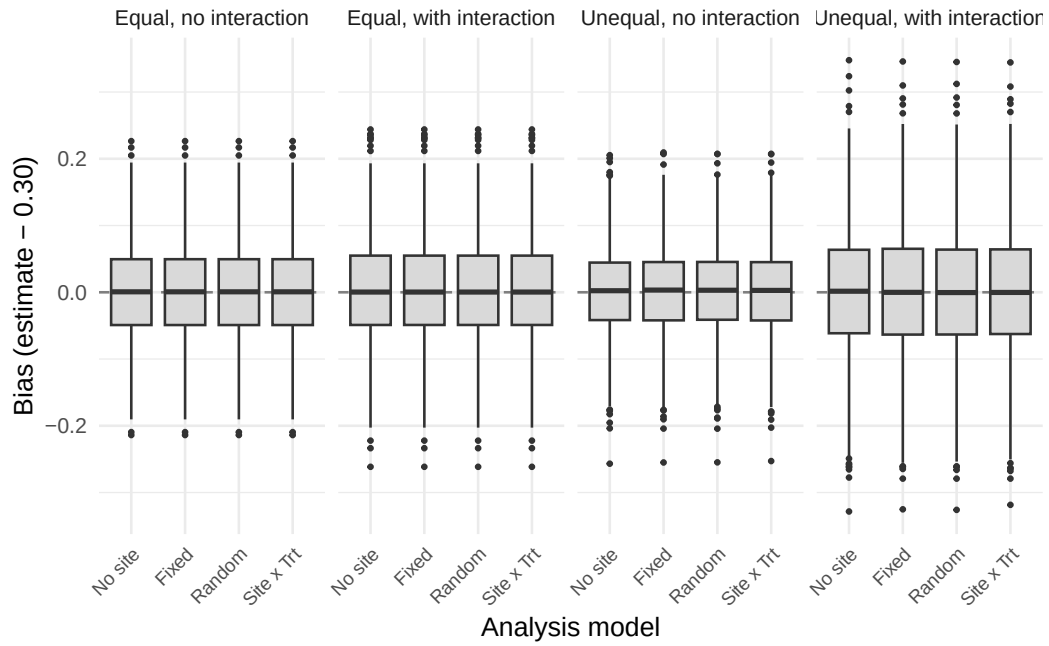


Figure 1: Bias of the treatment effect estimate across four models and four simulation scenarios.

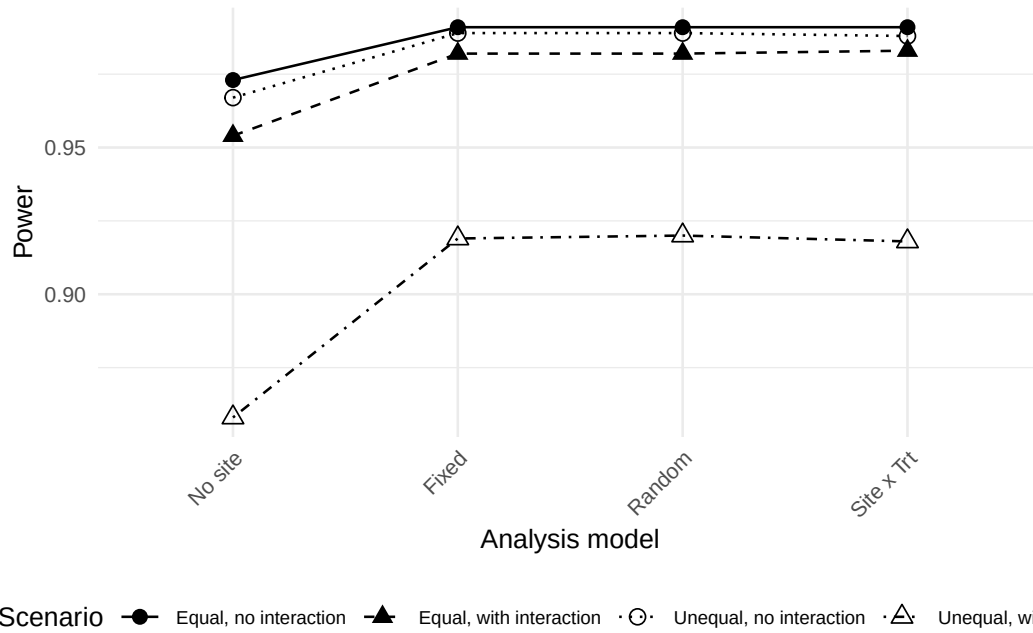


Figure 2: Power to detect the treatment effect across four models and four simulation scenarios.

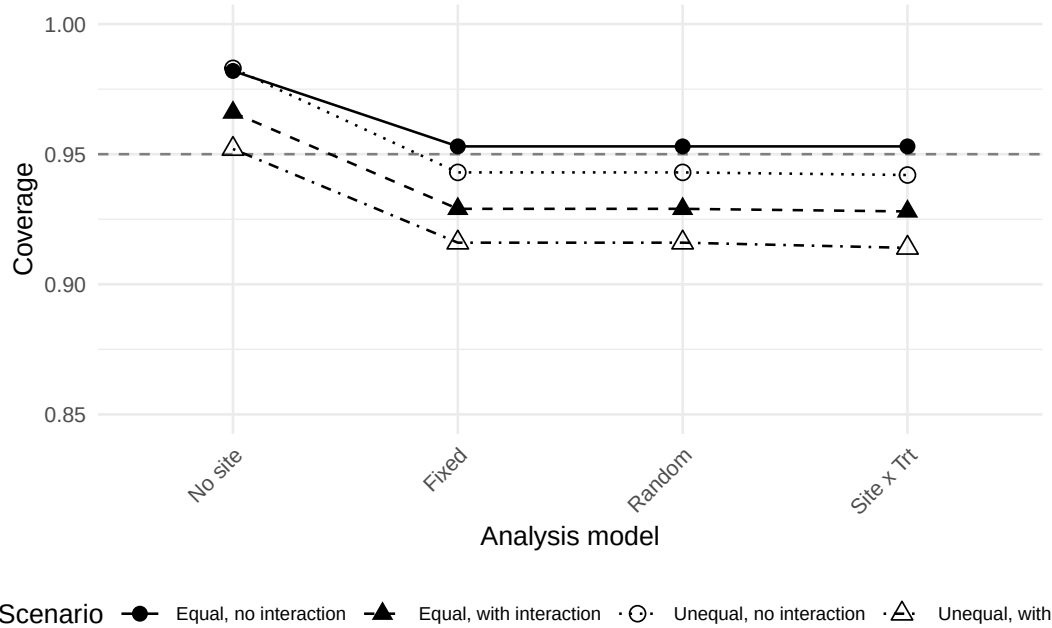


Figure 3: Coverage of nominal 95 percent confidence intervals across four models and four simulation scenarios. The dashed line indicates the nominal 95 percent level.

Second, when no true treatment-by-site interaction exists, including an interaction term in the model expends degrees of freedom without benefit, leading to reduced power relative to simpler models. This finding supports the common recommendation to include the interaction term only when there is a strong a priori reason to expect treatment effect heterogeneity across sites^{13,14}.

Third, the regulatory guidance from ICH E9, the FDA, and the EMA all converge on the recommendation that stratification variables should be included in the analysis model⁴⁻⁶. Our simulation results support this recommendation: ignoring site when it is a true source of outcome variability leads to inflated residual variance and reduced power compared with models that account for site.

Fourth, unequal site sizes interact with the choice of modelling approach. The random-effects model naturally handles imbalanced designs by borrowing information across sites, whereas the fixed-effects model treats each site independently and can perform poorly when some sites contribute sparse data^{3,14}.

Several limitations warrant acknowledgment. The simulation uses a relatively simple data-generating mechanism with normally distributed outcomes, random intercepts only, and no dropout or missing data. Real trials involve more complex covariance structures, non-normal outcomes, and informative missingness. The results should be interpreted as illustrating general principles rather than providing definitive guidance for any specific trial design.

234 6 Future research

235 Several directions merit further investigation:

- 236 1. **Non-normal outcomes.** Extending the simulation to binary and time-to-event end-
237 points would complement the present results for continuous outcomes and address the
238 findings of¹¹ and¹² regarding sparse events per site.
- 239 2. **Complex covariance structures.** The present simulation uses compound symmetry
240 (random intercept only) for both site and subject effects. Examining unstructured or
241 autoregressive within-subject correlations, and allowing random slopes at the site level,
242 would better approximate real longitudinal trial data.
- 243 3. **Informative site sizes.** When site size is correlated with patient characteristics or
244 treatment effect, ignoring this informativeness may bias results. Methods that model
245 the relationship between site size and outcome deserve attention.
- 246 4. **Adaptive designs.** In platform trials and other adaptive designs, sites may be added
247 or dropped during the trial. The implications for site-effect adjustment in these settings
248 are not well understood.
- 249 5. **Software tools.** Development of R functions or Shiny applications that allow trialists
250 to explore the impact of different site-adjustment strategies on their specific trial design
251 could facilitate informed pre-specification of analysis plans.
- 252 6. **Bayesian approaches.** Bayesian hierarchical models offer a natural framework for
253 modelling site effects and treatment-by-site heterogeneity, with the additional advantage
254 of incorporating prior information. Comparing frequentist and Bayesian approaches in
255 the multicenter trial context would be valuable.
- 256 7. **Empirical review.** A systematic review of how site effects are handled in recently pub-
257 lished trials in major medical journals (NEJM, JAMA, Lancet) would quantify current
258 practice and identify gaps between methodological recommendations and applied work,
259 extending the review by
260 1.

261 7 Conflict of interest

262 The author declares no conflicts of interest.

263 8 Funding

264 This work received no specific funding.

265 9 Data availability statement

266 The R simulation engine, driver scripts, and cached results that populate the manuscript's
267 tables and figures are openly available in the `sitecovariate` project repository. Specific
268 paths are listed in the Reproducibility section below.

269 10 Reproducibility

270 This manuscript was rendered with R 4.4.x. Principal packages used by the simulation pipeline
271 are `lme4` (the linear-mixed fitter), `MASS` (multivariate normal data generation), `furrr` and
272 `future` (parallel processing), `dplyr` and `ggplot2` (reporting), and `rmarkdown` plus `bookdown`
273 for the manuscript build.

274 **Random-number generator.** The simulation driver follows¹⁷ §4.1: `RNGkind` is pinned and
275 `set.seed(20260411)` is called once at the start of the program. Per-replicate seeds derive
276 from this master stream.

277 **Data and code availability.** The simulation engine is at `analysis/scripts/sim_site_covariate.R`.
278 Cached simulation outputs are at `analysis/data/`. The manuscript source, bibliography
279 (`references.bib`), and SIM CSL file are at `analysis/report/`.

280 11 References

281 11.1 Morris et al. (2019) ADEMP Compliance

282 We audited this simulation study against the reporting standards proposed by Morris, White,
283 and Crowther¹⁷. The full audit is recorded at `docs/morris-audit-2026-04-17.md`.

284 **Verdict:** Partially compliant.

285 Key gaps identified:

- 286 • `set.seed()` inside the worker function is reseeded on every scenario call (4 scenarios, 4
287 reseeds).
- 288 • No Monte Carlo SE on any metric (`summarise_sim()`).
- 289 • `n_sim = 1000` is hardcoded without a target-MCSE derivation.

290 A remediation plan is recorded in the audit file and will be taken up in a subsequent revision.

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